Additional Problems

1. ***This problem provides an instance with finite lead time with an additional twist where time-dependent costs and the demand rate are given in months. I forgot to move this to the Additional Problems section after finite lead time discussion.*** A local grocery store in Gainesville sells an average of 15 boxes of a certain brand and type of pasta every day. The store manager wants to decide on how much of this product to order so that she minimizes the average monthly cost of order and inventory holding. Placing an order with the distribution center costs $10 per order. The manager estimates that the inventory holding cost for this item is $0.5/box/month. Assume that a month is 30 days.
    a. Determine how much and how frequently, the manager should order this type of pasta. The order size does not have to be an integer value. State the order and inventory holding costs per month explicitly under the ordering policy you suggest.
    b. Suppose now that it takes 2 days for the distribution center to deliver an order to the store after the manager places the order. Update your inventory ordering policy accordingly. The reorder level does not have to be an integer value.
    c. How would your answer to part (b) above change, if it takes 12 days for the distribution center to deliver the order?

a. Note that λ below represents the monthly demand rate!

$$q^* = \sqrt{\frac{2K\lambda}{h}} = \sqrt{\frac{2(10)(30)(15)}{(0.5)}} = 134.16 \text{ boxes}$$

$$T^* = \frac{q^*}{\lambda} = \frac{134.16}{(30)(15)} = 0.2981 \text{ months} = 8.943 \text{ days}$$

Optimal policy: Order 134.16 boxes of pasta every 8.943 days with a total monthly cost of

$$C(q^*) = \frac{(10)(450)}{134.16} + \frac{(0.5)(134.16)}{2} = 33.54 + 33.54 = \$67.08/\text{year}$$

b. The lead time is 2 days. The demand during lead time is $(2)(15) = 30$. We have the following

$$r = \text{mod}(30,134.16) = 30 \text{ units}$$

Optimal policy: Order 134.16 boxes of pasta when the on-hand inventory level drops down to 30 units.

c. The lead time is 12 days. The demand during lead time is $(12)(15) = 180$. We have the following

$$r = \text{mod}(180,134.16) = 45.84 \text{ units}$$

Optimal policy: Order 134.16 boxes of pasta when the on-hand inventory level drops down to 45.84 units.

2. **This is a standard EPQ example, which I forgot to move to the Additional Problems section after EPQ.** Clara's is a family-owned manufacturer of organic lip balms and hand lotions. Clara's produces 5,000 lip balms a year and it costs $2 to produce a lip balm. It is estimated that the cost of set up to produce lip balms is $50 (the residue from the hand lotion production must be removed from the production line before starting the production of a lip balm batch). Inventory holding cost is based on a 12 percent annual interest rate. The demand for lip balms is 2,400 sticks per year.
    a. What is the optimal batch size of a production run?
    b. What is the length of a production cycle?
    c. What is the average annual cost of production setup and inventory holding for lip balms?

a.

$$q^* = \sqrt{\frac{2K\lambda}{h\left(1-\frac{\lambda}{P}\right)}} = \sqrt{\frac{2(50)(2{,}400)}{(0.12)(2)\left(1-\frac{2{,}400}{5{,}000}\right)}} = 1{,}386.75 \text{ units}$$

b.

$$T_1 = \frac{q^*}{P} = \frac{1{,}386.75}{5{,}000} = 0.2774 \text{ years}$$

c.

$$C(q^*) = \frac{(50)(2{,}400)}{1{,}386.75} + \frac{(0.12)(2)\left(1-\frac{2{,}400}{5{,}000}\right)(1{,}386.75)}{2} = 86.53 + 86.53 = \$173.06/\text{year}$$

3. ***This was an exam question in the past and is a modification of the standard all-units discount.*** One of the ways of cutting carbon dioxide (CO_2) emissions is by recycling of materials such as steel, aluminum, and copper. Earthworks Recycling Company buys and sells several recyclable materials, including aluminum in various forms (cans, cast metal, extrusions, sheets, car wheels and rims, etc.). Earthworks Recycling Company uses a *reverse* all-units discount scheme for the aluminum cans they buy as given below:

Order size (lbs)	Price ($/lb)
$0 < q \leq 50$	30.0
$50 < q \leq 100$	32.5
$100 < q$	35.0

The amount of aluminum cans purchased from a large collector company in the vicinity is about 10,000 pounds annually. The fixed order cost is $5 per order, and the inventory holding rate is based on an annual inventory carrying rate of 25 percent. What is the optimal order quantity? Please justify your answer clearly.

We have the following:

$$q^*_{c=30} = \sqrt{\frac{2K\lambda}{ic_{30}}} = \sqrt{\frac{2(5)(10{,}000)}{(0.25)(30)}} = 115.47 \text{ lbs (Not Realizable)}$$

$$q^*_{c=32.5} = \sqrt{\frac{2K\lambda}{ic_{32.5}}} = \sqrt{\frac{2(5)(10{,}000)}{(0.25)(32.5)}} = 110.94 \text{ lbs (Not Realizable)}$$

$$q^*_{c=35} = \sqrt{\frac{2K\lambda}{ic_{35}}} = \sqrt{\frac{2(5)(10{,}000)}{(0.25)(35)}} = 106.91 \text{ lbs (Realizable)}$$

Note that there is only one realizable EOQ level. Since we are using a reverse all-units discount scheme, the minimum average total cost solution will occur either a the smallest realizable EOQ level or at a break point below it. Hence, we need to compare the average annual total cost at $q^*_{c=35}$ as well as at the breakpoints of b_1 = 50lbs and b_2 = 100lbs. We have the following:

$$q^*_{c=35} = 106.91 \text{ lbs}$$

$$C(q^*_{c=35}) = (35)(10{,}000) + \frac{(5)(10{,}000)}{106.91} + \frac{(0.25)(35)(106.91)}{2} = \$350{,}935.41\ /years$$

$$q_{b_2} = 100 \text{ lbs}$$

$$C(q_{b_2}) = (32.5)(10{,}000) + \frac{(5)(10{,}000)}{100} + \frac{(0.25)(32.5)(100)}{2} = \$325{,}906.30\ /years$$

$$q_{b_1} = 50 \text{ lbs}$$

$$C(q_{b_1}) = (30)(10{,}000) + \frac{(5)(10{,}000)}{50} + \frac{(0.25)(30)(50)}{2} = \$301{,}187.50\ /years$$

In summary, it would be optimal to purchase recyclable aluminum cans from the collector using an order size of 50lbs every 50/10,000=0.005 years.

Intuition: *Recall how the overall average total cost function could be plotted when all-units discount is in effect.* ***Figure 1*** *below shows how the overall average total cost function could be plotted when reverse all-units discount is in effect.*

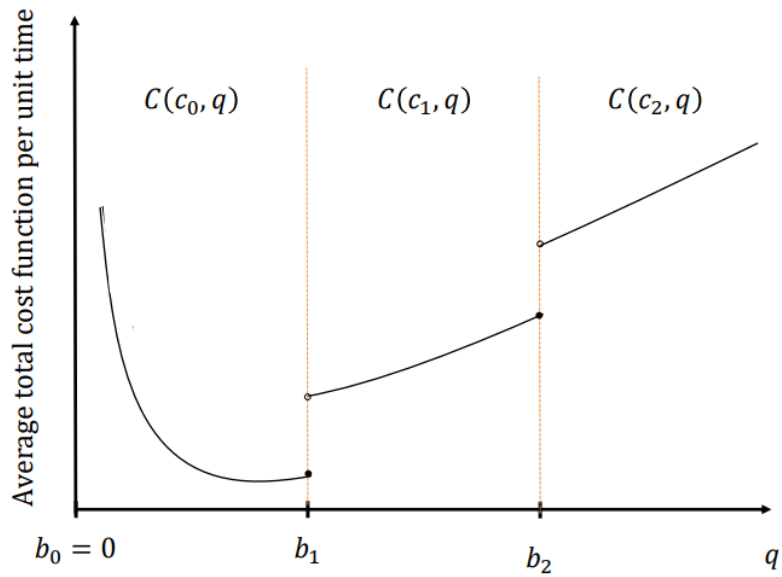

Figure 1. Average total cost function when a reverse all-units discount is in effect, i.e., when the unit cost increases as order quantity increases. Note that this image is not representative of the cost curves in this example but is intended to offer you some general insight for reverse all-units.

Note: *If a reverse incremental unit discount were to be conceptualized, note that the standard approach used to address the incremental discount scheme can be applied as is.*

4. You work for a bottling company and your team has been tasked with presenting the candidate solutions for a new soda bottle based on the supplier's all-units discount scheme as illustrated below. Your coworker solved the equation $q_i^{EOQ} = \sqrt{\frac{2K\lambda}{ic_i}}$ for each price and wants to present those quantities as the four definitive candidates.

Order size (unit)	Price ($/unit)	Economic Order Quantity
$0 \leq q < 10{,}000$	0.30	$q_1^{EOQ} = 23{,}458$
$10{,}000 \leq q < 20{,}000$	0.28	$q_2^{EOQ} = 27{,}302$
$20{,}000 \leq q < 50{,}000$	0.25	$q_3^{EOQ} = 34{,}120$
$50{,}000 \leq q$	0.20	$q_4^{EOQ} = 45{,}791$

Are these quantities the candidate solutions? If not, find the appropriate list of candidate solutions based on the EOQ quantities and breakpoints.

SOLUTION: These are not the candidate solutions, we need to evaluate each case and adjust the economic order quantity depending on what case it falls in. The following cases account for all-units discount schemes

a. Case 1: If $q_j^{EOQ} < b_j$, then $q_j^* = b_j$.

b. Case 2: If $b_j \leq q_j^{EOQ} < b_{j+1}$, then $q_j^* = q_j^{EOQ}$.

c. Case 3: If $b_{j+1} < q_j^{EOQ}$, then do nothing (i.e., there is no candidate solution).

Order size (unit)	Economic Order Quantity	Case	Candidate Solution
$0 \leq q < 10{,}000$	$q_1^{EOQ} = 23{,}458$	Case 3	None
$10{,}000 \leq q < 20{,}000$	$q_2^{EOQ} = 27{,}302$	Case 3	None
$20{,}000 \leq q < 50{,}000$	$q_3^{EOQ} = 34{,}120$	Case 2	34,120
$50{,}000 \leq q$	$q_4^{EOQ} = 45{,}791$	Case 1	50,000

Therefore, the true candidate solutions are $q_3^* = 34{,}120 \; and \; q_4^* = 50{,}000$.

5. Consider a scenario with two different suppliers offering different discount schemes. Supplier A offers an all-unit discount scheme with a unit price of $10.00 for quantities up to (but excluding) 5,000 and $9.00 if the quantity is 5,000 or higher. Supplier B offers an incremental discount scheme with a unit price of $12.00 for quantities up to 1,000, unit price of $10.00 for any quantity between 1,000 and 3,000, and unit price of $7.50 for any quantity 3,000 or higher.

The product's demand is 1,000 units/week The cost of ordering from either supplier is $1,000 and an annual inventory carrying rate of 25 percent is used by the company. The company operates for 52 weeks a year.

a. What are the possible candidate quantities for supplier A's all-unit discount scheme?
b. What are the possible candidate quantities for supplier B's incremental discount scheme?
c. Which inventory policy is optimal? Your answer should include the choice of supplier and the optimal quantity to order.

SOLUTION:

a. We use the standard EOQ formula to find the possible economic order quantities for each breakpoint. We will then filter out any solutions that are not realizable.

$$q_{c=10}^* = \sqrt{\frac{2K\lambda}{ic_{10}}} = \sqrt{\frac{2(1{,}000)(1{,}000)(52)}{(0.25)(10)}} = 6{,}449.81 \; units(\text{Not Realizable})$$

$$q_{c=9}^* = \sqrt{\frac{2K\lambda}{ic_9}} = \sqrt{\frac{2(1{,}000)(1{,}000)(52)}{(0.25)(9)}} = 6{,}798.69 \; units \; (\text{Realizable})$$

Since 6,449.81 > 5,000, that economic order quantity is not a candidate solution. 6,798 is greater than 5,000 and is the only candidate solution for scenario A.

b. Since this scenario uses an incremental discount scheme, we will use the appropriate EOQ equation to account for this. First, we find the previous cost from each breakpoint.

$$V_0 = 0$$
$$V_1 = 12(1{,}000) = 12{,}000$$
$$V_2 = 12(1{,}000) + (10)(3{,}000 - 1{,}000) = 32{,}000$$

We now use these values to calculate the economic order quantity for each range

$$q_0^{EOQ} = \sqrt{\frac{2(1{,}000)(52)\big(1{,}000 + 0 - (0)(12)\big)}{(0.25)(12)}} \;=\; 5{,}887.84\ units$$

$$q_1^{EOQ} = \sqrt{\frac{2(1{,}000)(52)\big(1{,}000 + 12{,}000 - (1{,}000)(10)\big)}{(0.25)(10)}} \;=\; 11{,}171.84\ units$$

$$q_2^{EOQ} = \sqrt{\frac{2(1{,}000)(52)\big(1{,}000 + 32{,}000 - (3{,}000)(7.5)\big)}{(0.25)(7.5)}} \;=\; 24{,}132.97\ units$$

Since both q_0^{EOQ} and q_1^{EOQ} are greater than their breakpoints, we set $q_0^* = 1{,}000$ and $q_1^* = 3{,}000$ and leave $q_2^* = 24{,}132.97$ as it lies inside its breakpoints.

c. For part c, we need to compare the cost of our four total candidate solutions and select the cheapest one. We now evaluate the cost associated with each candidate solution starting with scenario A.

Scenario A Cost:

$$C_0(q_{c=9}^*) = 9(52{,}000) + \frac{1{,}000(52{,}000)}{6{,}798.69} + \frac{0.25(9)(6{,}798.69)}{2}$$
$$= 468{,}000 + 7648.53 + 7648.53 = \$483{,}297.05$$

Scenario B Costs

$$C_0(q_0^*) = \frac{(1{,}000)(52{,}000)}{1{,}000} + \left(\frac{0.25}{2} + \frac{52{,}000}{1{,}000}\right)[0 + (1000 - 0)12]$$
$$C_0(q_0^*) = \ 52{,}000 + 625{,}500 = \$677{,}500$$
$$C_1(q_1^*) = \frac{(1{,}000)(52{,}000)}{3{,}000} + \left(\frac{0.25}{2} + \frac{52{,}000}{3{,}000}\right)[12{,}000 + (3{,}000 - 1{,}000)10]$$
$$C_1(q_1^*) = \ 17{,}333.33 + 558{,}666.67 = \$576{,}000$$
$$C_2(q_2^*) = \frac{(1{,}000)(52{,}000)}{24{,}132.97} + \left(\frac{0.25}{2} + \frac{52{,}000}{124{,}132.97}\right)[32{,}000 + (24{,}132.97 - 3{,}000)7.5]$$
$$C_2(q_2^*) = \$2{,}154.73 + \$434{,}333.80 = \$436{,}453.32/\text{year}$$

The optimal policy would be to order from Supplier B using an order size of 24,132.97 units.

6. New semester! You just moved into a new apartment and are thinking of changing some of your purchasing habits. You normally buy each of your bathroom products individually when you run out of them. Your demand, price, and order cost (cost of driving to the nearest store) for each product are as follows

Product	Demand	Price	Order cost
Body Wash	1 bottle per month	$15.00	$1.00

Toothpaste	1 tube every 2 months	$6.00	$2.00
Toilet Paper	1 roll per week	$2.00	$1.00

For each product, you have a 10% inventory carrying rate. You are thinking of going to a different store that has everything you need, the joint total cost for making this trip would be $2.50. Would it be cheaper to buy everything jointly or separately?

SOLUTION: We can answer this question by splitting our solution into the two different scenarios, separate and joint. Keep in mind the different units with each demand, as products are bought at different points normally.

Scenario 1: Separate: We must first find the economic order quantity for each product separately and then use the quantities and sum the total cost across each product.

$$q_{BW}^* = \sqrt{\frac{2(1)(1)(12)}{(0.10)(15)}} = 4\ units$$

$$q_{Tooth}^* = \sqrt{\frac{2(2)(1)(6)}{(0.10)(6)}} = 6.32\ units$$

$$q_{Toilet}^* = \sqrt{\frac{2(1)(1)(52)}{(0.10)(2)}} = 22.8\ units$$

Now that we have found the economic order quantities for each product, we must find the cost associated with each.

$$C(q_{BW}^*) = (15)(1)(12) + \frac{(1)(1)(12)}{4} + \frac{(0.10)(15)(4)}{2} = 180 + 3 + 3 = \$186$$

$$C(q_{Tooth}^*) = (6)(1)(6) + \frac{(2)(1)(6)}{6.32} + \frac{(0.10)(6)(6.32)}{2} = 36 + 1.90 + 1.90 = \$39.8$$

$$C(q_{Toilet}^*) = (2)(1)(52) + \frac{(1)(1)(52)}{22.8} + \frac{(0.10)(2)(22.8)}{2} = 104 + 2.28 + 2.28 = \$108.56$$

$$C(q_{BW}^*, q_{Tooth}^*, q_{Toilet}^*) = 186 + 39.8 + 108.56 = \$334.36$$

Scenario 2: Jointly: We now evaluate the case of jointly ordering by first finding the optimal cycle length and then using that to find the total cost.

$$T^* = \sqrt{\frac{2(2.5)}{(0.10)(15)(1)(12) + (0.10)(6)(1)(6) + (0.10)(2)(1)(52)}} = \sqrt{\frac{5}{18 + 3.6 + 10.4}} = 0.3953$$

$$C_J(T^*) = \sum_{j=1}^{n} c_j\lambda_j + \sqrt{2K\sum_{j=1}^{n} h_j\lambda_j}$$

$$= 15(12) + 6(6) + 2(52) + \sqrt{2(2.5)((1.5)(12) + (0.6)(6) + (0.2)(52))} = 320 + 12.65 = \$332.65$$

In conclusion, joint ordering is cheaper than ordering each product separately. The separate ordering policy costs $334.36 and would require you to travel every 0.25 years to purchase supplies. The joint ordering policy costs $332.65 and requires you to travel every 0.3953 years to purchase supplies.